HIGH-LEVEL DESIGN (HLD) DOCUMENT

AI-Based Monitoring Solution for Oil & Gas Industry

Document Version: 1.0 **Date**: December 2025 **Project Code**: OG-AI-MON-2025 **Classification**: Confidential

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1. DOCUMENT OVERVIEW

1.1 Purpose

This High-Level Design (HLD) document provides the architectural blueprint for the AI-Based Monitoring Solution for Oil & Gas operations. It describes the overall system architecture, major components, data flows, and integration patterns without diving into implementation-level details.

1.2 Scope

In Scope	Out of Scope
System architecture overview	Detailed code implementation
Component interactions	Database schema design
Technology stack selection	API specifications
Integration patterns	Configuration parameters
Security architecture	Deployment scripts
Network topology	Unit test cases

1.3 Audience

- Solution Architects
- Technical Leads
- IT Directors
- Project Managers
- Vendor Partners

1.4 Referenced Documents

Document	Version	Description
BRD	1.0	Business Requirements Document
Security Policy	2.1	Corporate IT Security Guidelines
OT Standards	3.0	Operations Technology Standards

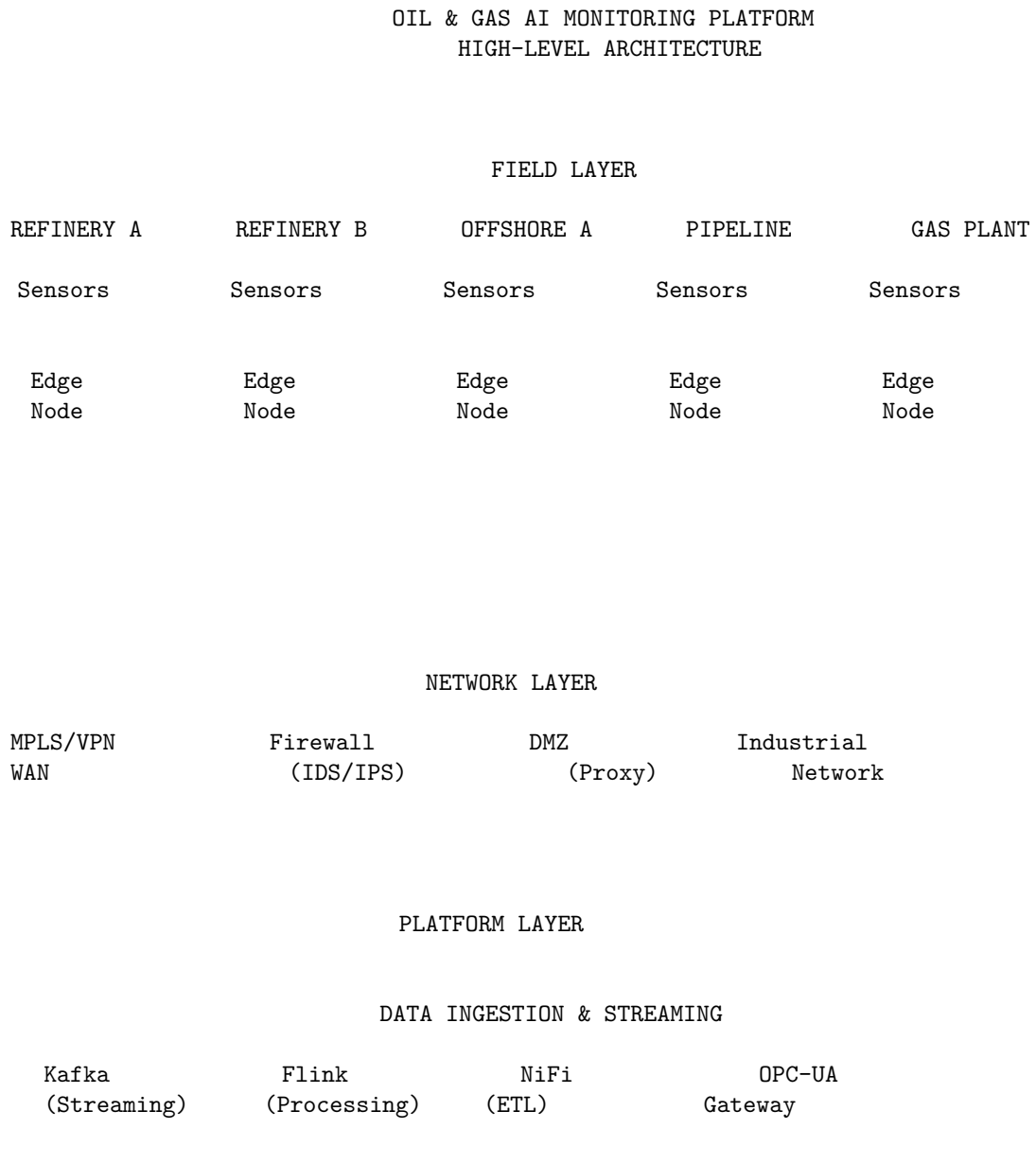
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2. ARCHITECTURE OVERVIEW

2.1 Architecture Principles

Principle	Description	Rationale
Edge-First	Process data at edge, send insights to cloud	Reduce latency, bandwidth, and cloud costs
Fail-Safe	AI layer can fail without impacting operations	Safety-critical systems must remain operational
Modular	Loosely coupled microservices	Independent deployment and scaling
Secure by Design	Zero-trust, defense in depth	Protect critical infrastructure
Hybrid Deployment	Edge + on-premise private cloud	Meet data sovereignty requirements
Vendor Agnostic	Open standards and APIs	Avoid lock-in, enable best-of-breed

2.2 High-Level Architecture Diagram



DATA LAYER

InfluxDB (TimeSeries)	PostgreSQL (Metadata)	Milvus (Vector DB)	S3/MinIO (Objects)

AI/ML LAYER

Predictive Maintenance (ML)	Anomaly Detection (DL)	Computer Vision (CV)	RAG/LLM Assistant (NLP)

SERVICE LAYER

Alert Manager	Work Order Service	Report Generator	API Gateway

APPLICATION LAYER

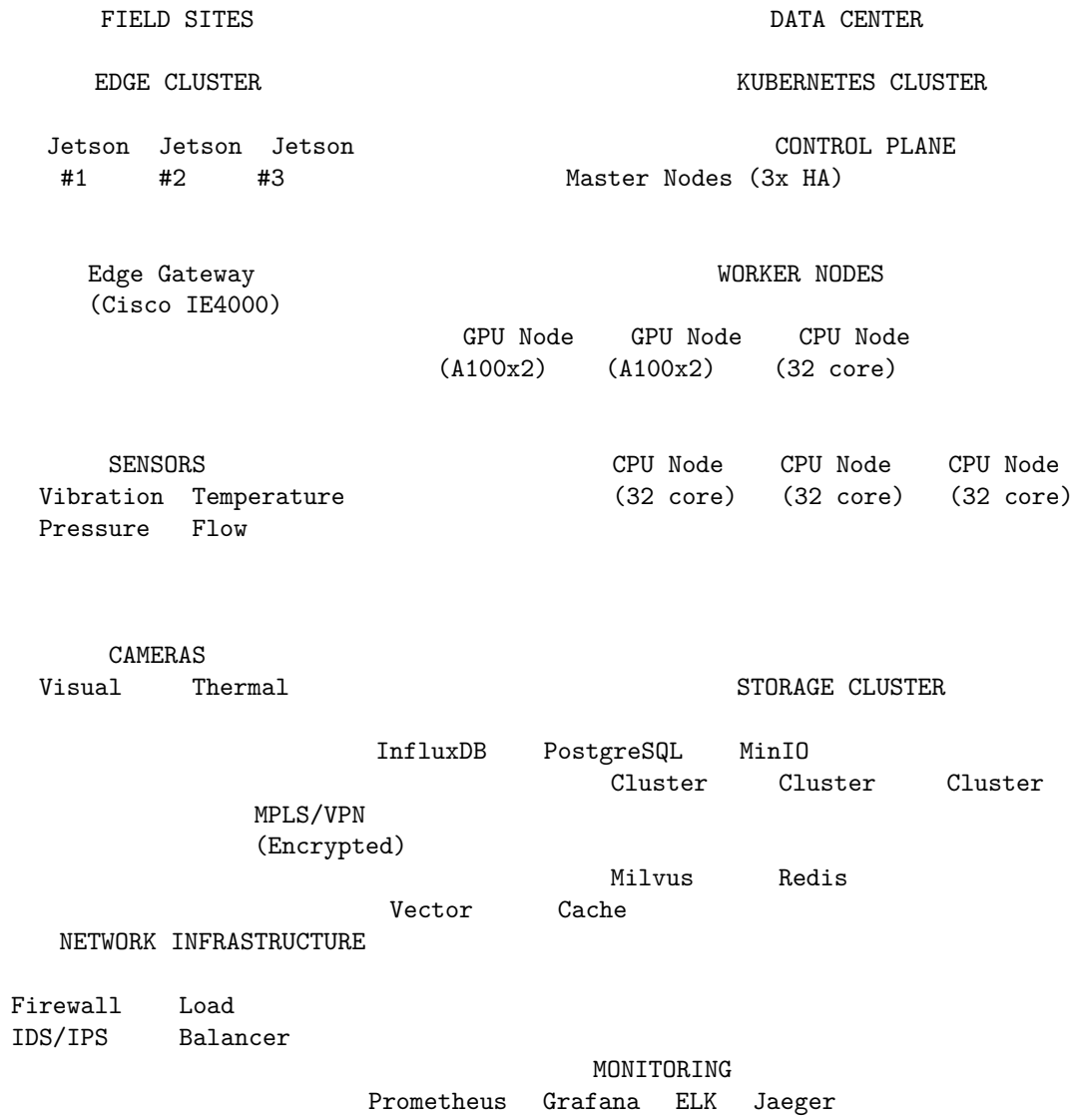
Control Room Dashboard	Mobile App (iOS/And)	RAG Chatbot (Web)	Executive Dashboard (BI)	Admin Portal (Web)

INTEGRATION LAYER

SAP PM	Maximo	OSIsoft PI	DCS/SCADA	Email/SMS

2.3 Deployment View

DEPLOYMENT ARCHITECTURE

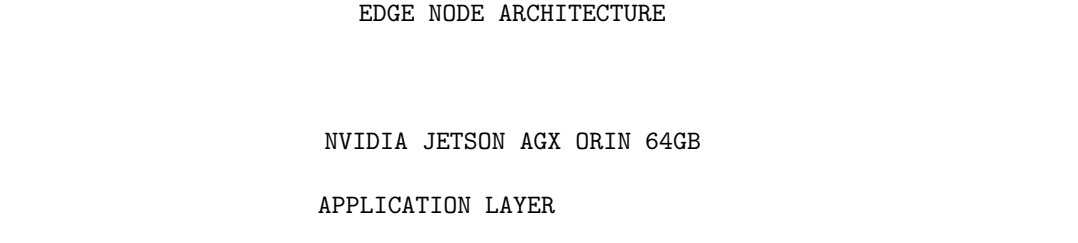


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3. COMPONENT ARCHITECTURE

3.1 Edge Computing Layer

3.1.1 Edge Node Architecture



```
Inference       Video           Data            MQTT
Engine          Pipeline        Collector       Client
(TensorRT)      (DeepStream)    (Python)        (Paho)


                    RUNTIME LAYER

Docker          CUDA 12.x       TensorRT
Runtime         Runtime         8.6


                      OS LAYER
Ubuntu 22.04 LTS + JetPack 6.0


                   I/O INTERFACES

GigE            USB 3.2         CAN Bus         GPIO
(4 ports)       (4 ports)       (2 ports)       (40 pin)
```

CONNECTED DEVICES:

- 4x Industrial Cameras (GigE Vision)
- 1x DAQ Module (USB - NI-9234)
- 1x Industrial Gateway (Ethernet - OPC-UA)
- 1x Thermal Camera (USB)

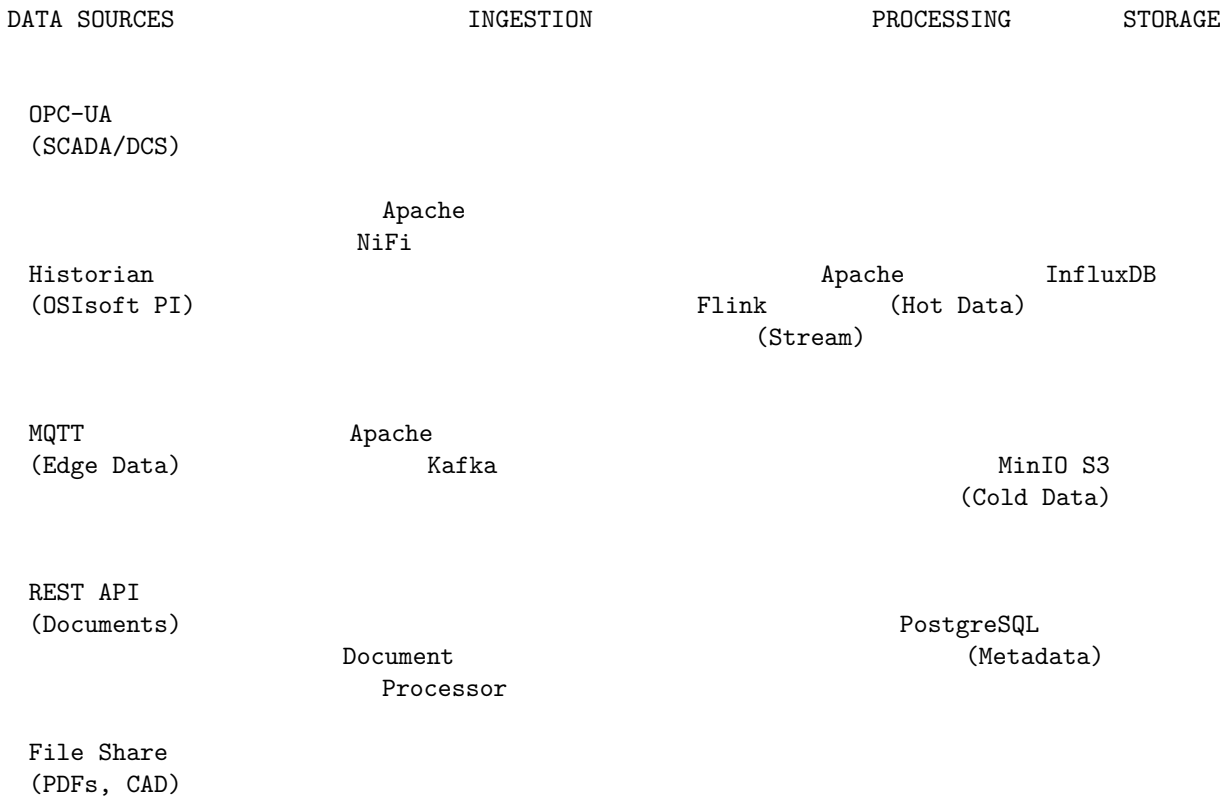
3.1.2 Edge Functions

Function	Description	Latency Target	Processing
Vibration Analysis	FFT, feature extraction	<10ms	Real-time
Anomaly Detection	Autoencoder inference	<50ms	Real-time
Object Detection	YOLO inference	<33ms (30 fps)	Real-time
Data Aggregation	1-minute rollups	N/A	Batch
Store & Forward	Buffer during disconnection	N/A	Async

3.2 Data Ingestion Layer

3.2.1 Data Flow Architecture

```
                    DATA INGESTION FLOW
```



Data Volumes:

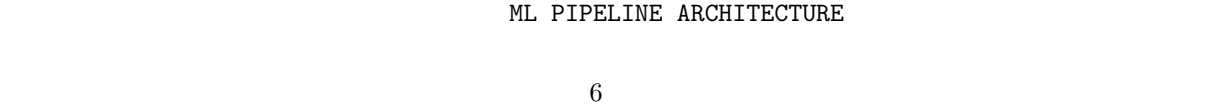
- Sensor Data: 1 TB/day (1000 sensors @ 1 Hz average)
- Video: 10 TB/day (50 cameras @ 4K @ 15fps compressed)
- Documents: 50 GB initial + 1 GB/month

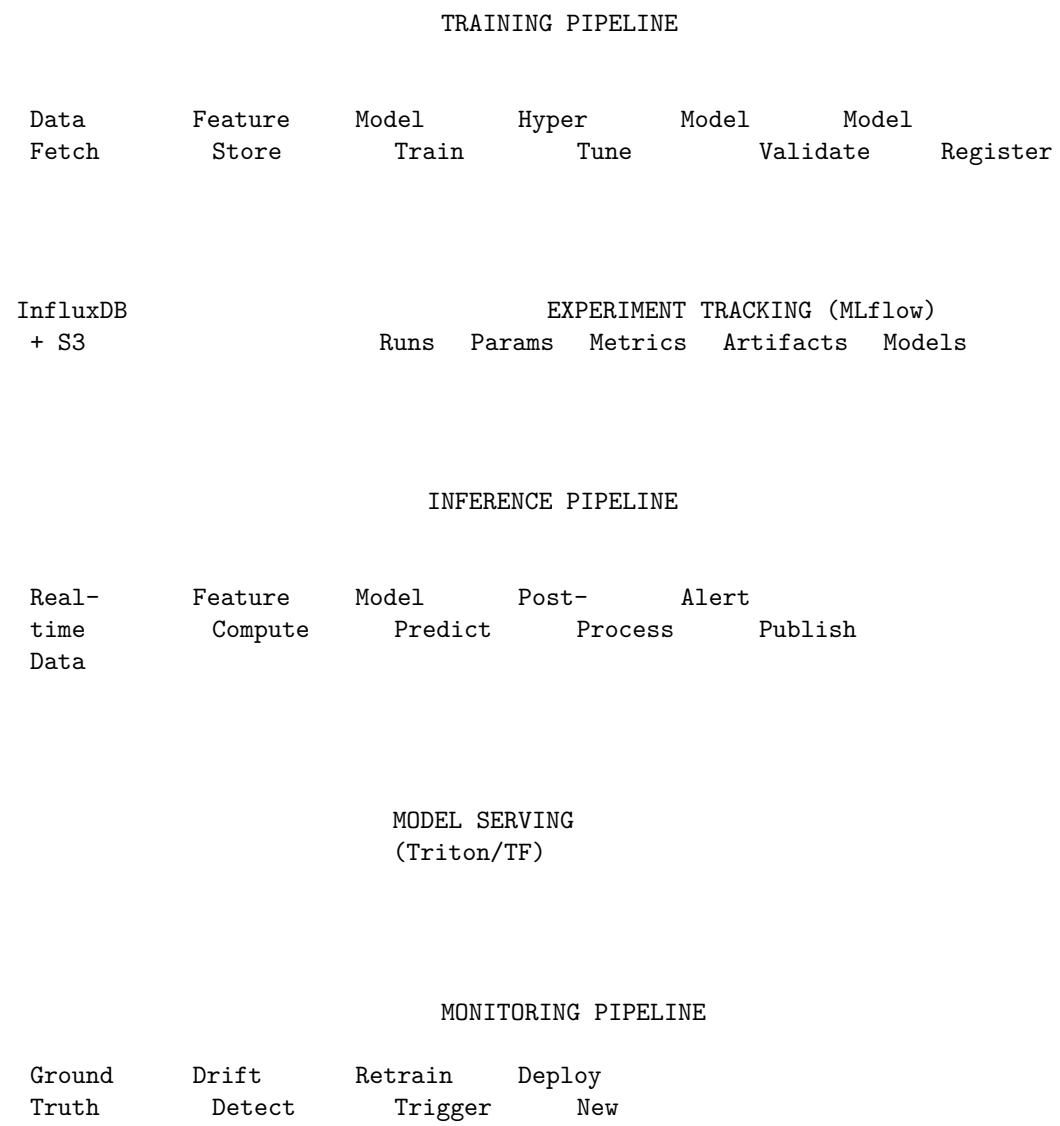
3.2.2 Kafka Topic Structure

Topic	Partitions	Retention	Description
sensor.raw.vibration	20	7 days	Raw vibration data
sensor.raw.temperature	10	7 days	Temperature readings
sensor.raw.pressure	10	7 days	Pressure readings
sensor.processed.features	20	30 days	Extracted features
alerts.anomaly	5	90 days	Anomaly detections
alerts.prediction	5	90 days	Failure predictions
vision.detections	10	7 days	CV detection results
documents.chunks	5	Forever	Document embeddings

3.3 AI/ML Layer

3.3.1 ML Pipeline Architecture





3.3.2 Model Catalog

Model	Type	Input	Output	Deployment	Update Frequency
Vibration Predictor	XGBoost + LSTM	FFT features	RUL (days)	Edge + Cloud	Monthly
Anomaly Detector	Autoencoder	Sensor vector	Anomaly score	Edge	Weekly
Corrosion Detector	YOLOv8-seg	1920x1080 image	Segmentation mask	Edge	Monthly

Model	Type	Input	Output	Deployment	Update Frequency
PPE Detector	YOLOv8-det	1920x1080 image	Bounding boxes	Edge	Monthly
Gas Leak Detector	Multi-modal CNN	RGB + Thermal	Detection + location	Edge	Monthly
RAG Retriever	BGE-large	Text query	Top-K chunks	Cloud	As needed
RAG Generator	Llama 3.1 70B	Query + context	Answer	Cloud	Quarterly

3.4 RAG Architecture

```
                                RAG SYSTEM ARCHITECTURE


                            DOCUMENT INGESTION PIPELINE


  Source      Parser      Chunker     Embed       Index       Milvus
  (PDF,       (PyMuPDF,    (Semantic    (BGE-         Store        Vector
  CAD,DOC)     Unstruct     512 tok)     large)                    DB


Document Types: P&IDs, Equipment Manuals, SOPs, Incident Reports, Standards
Volume: 50,000 documents, ~5M chunks


                                QUERY PIPELINE


  User        Query       Vector      Rerank      Context     LLM
  Query        Embed        Search       (Cross-      Build        Generate
                            (Top 20)     Encoder)     (Top 5)

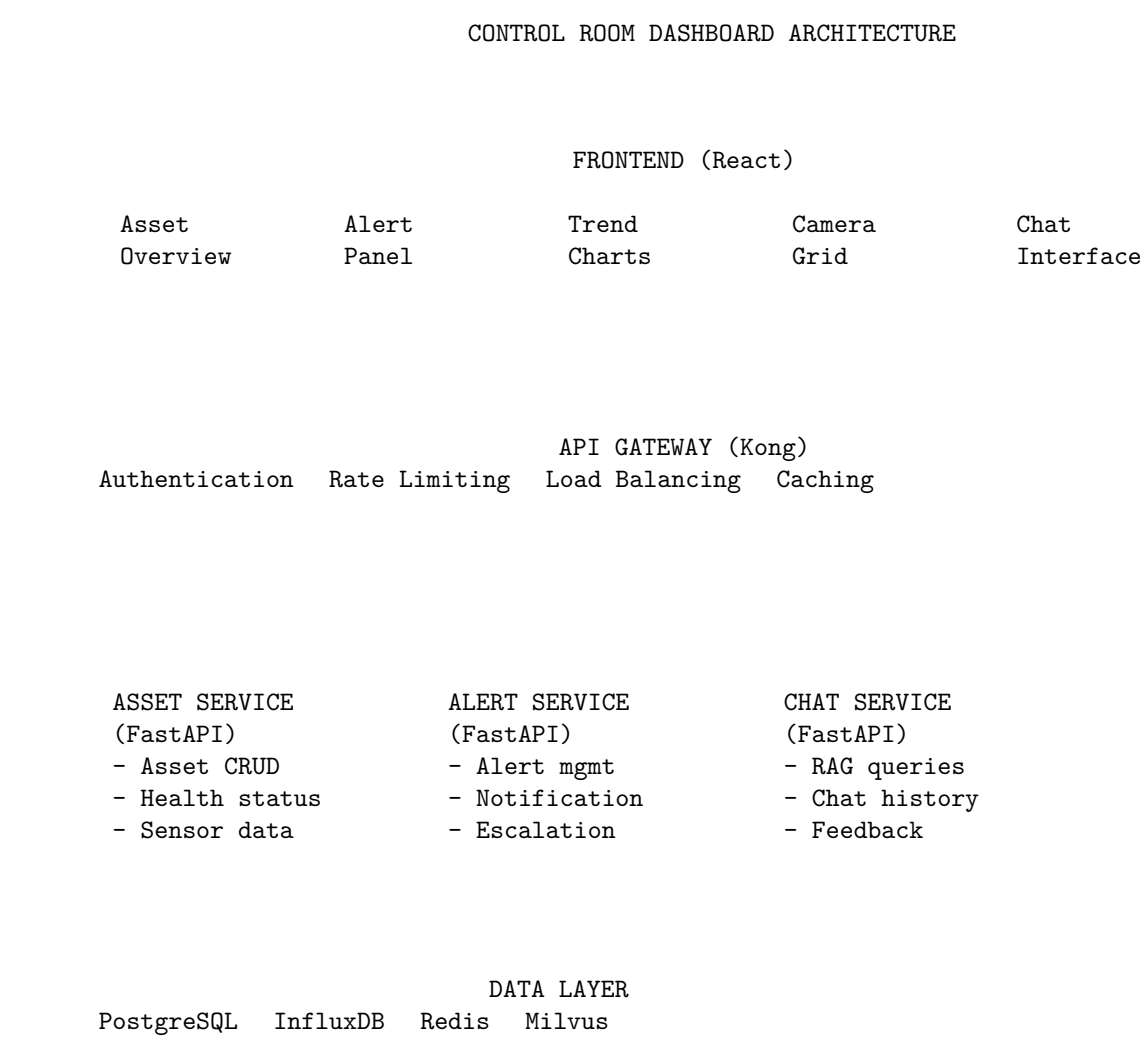

Latency: Query Embed (50ms) + Search (100ms) + Rerank (200ms) + Generate (2s) = ~2.5s


                             REAL-TIME IOT CONTEXT


  Current     Context     Inject      Included in LLM prompt
  Sensor       Format       into
  Values                    Prompt       "Current temperature: 85°C (WARNING)"
```

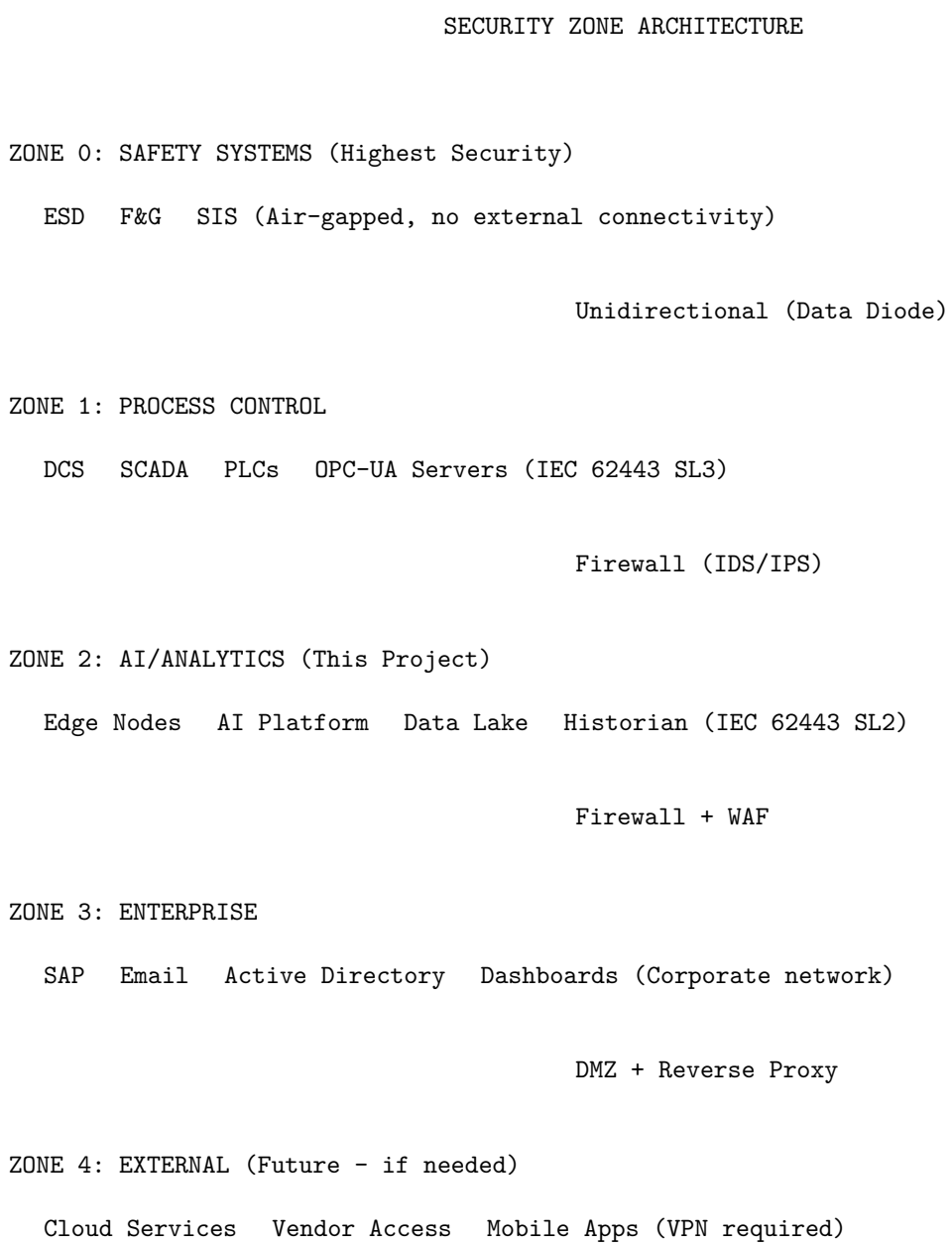
3.5 Application Architecture

3.5.1 Control Room Dashboard



4. SECURITY ARCHITECTURE

4.1 Security Zones



4.2 Security Controls

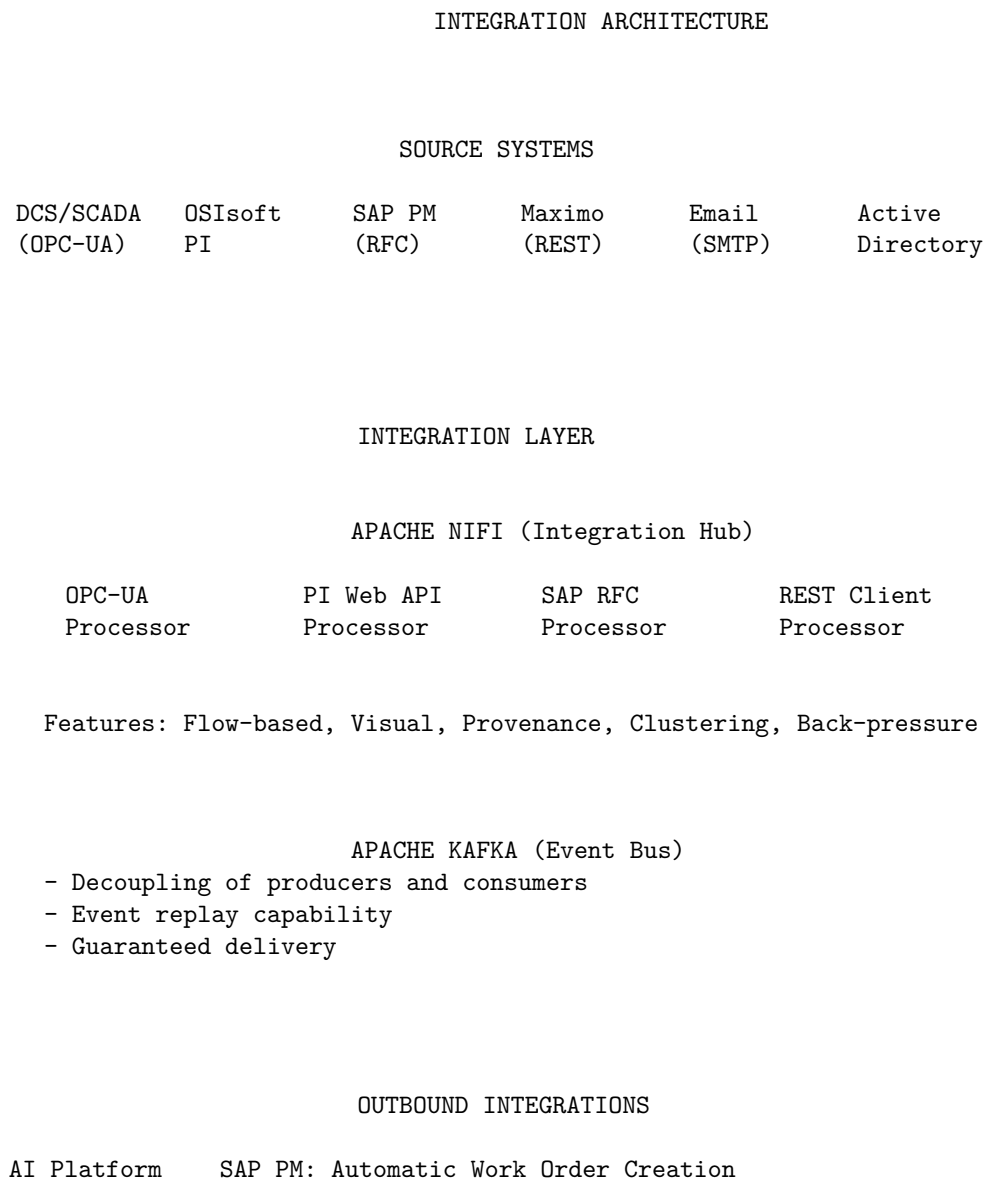
Control	Implementation	Standard
Authentication	Keycloak (OIDC/SAML)	OAuth 2.0

Control	Implementation	Standard
Authorization	RBAC with Keycloak	NIST 800-53
Encryption at Rest	AES-256	FIPS 140-2
Encryption in Transit	TLS 1.3	PCI-DSS
Network Segmentation	VLANs + Firewalls	IEC 62443
Intrusion Detection	Suricata IDS	NIST CSF
Logging & Audit	ELK Stack	SOC 2
Secrets Management	HashiCorp Vault	-

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5. INTEGRATION ARCHITECTURE

5.1 Integration Patterns



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AI Platform    Email/SMS: Alert Notifications
AI Platform    Historian: Prediction results for trending
AI Platform    Maximo: Asset health updates
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5.2 API Specifications Summary

API	Protocol	Format	Auth	Rate Limit
Asset API	REST	JSON	OAuth 2.0	1000/min
Sensor API	REST	JSON	OAuth 2.0	5000/min
Alert API	REST + WebSocket	JSON	OAuth 2.0	500/min
RAG API	REST	JSON	OAuth 2.0	100/min
SAP Integration	RFC/BAPI	XML	Basic	50/min

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6. SCALABILITY & PERFORMANCE

6.1 Capacity Planning

Component	Current	Year 1	Year 3	Year 5
Sensors	1,000	2,000	5,000	10,000
Cameras	50	100	200	500
Edge Nodes	20	40	100	200
Data Volume	1 TB/day	3 TB/day	10 TB/day	30 TB/day
Concurrent Users	50	100	200	500
LLM Queries	500/day	2,000/day	10,000/day	50,000/day

6.2 Performance Targets

Metric	Target	Measurement
Edge Inference Latency	<100ms	P99 latency
API Response Time	<500ms	P95 latency
Dashboard Load Time	<3s	P95 latency
RAG Query Response	<5s	P95 latency
Alert Processing	<10s	End-to-end
Data Ingestion Lag	<30s	Kafka consumer lag

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7. DISASTER RECOVERY

7.1 DR Architecture

Tier	Components	RPO	RTO	Strategy
Tier 1	Kubernetes Control Plane	0	15 min	Active-Active
Tier 2	Databases (PostgreSQL, InfluxDB)	1 hour	4 hours	Streaming Replication
Tier 3	Object Storage (MinIO)	4 hours	8 hours	Cross-site Replication
Tier 4	ML Models	24 hours	24 hours	S3 versioning

7.2 Backup Strategy

Data Type	Backup Frequency	Retention	Storage
PostgreSQL	Hourly incremental, Daily full	30 days	S3
InfluxDB	Continuous replication	7 years	DR site
MinIO	Daily sync	7 years	DR site
ML Models	On deployment	12 months	S3 + Git LFS
Configurations	On change	Forever	Git

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8. TECHNOLOGY SUMMARY

8.1 Technology Stack

Layer	Technology	Version	Purpose
Edge Hardware	NVIDIA Jetson AGX Orin	64GB	Edge AI inference
Edge Gateway	Cisco IE4000	-	Industrial networking
Container Platform	Kubernetes (OpenShift)	4.14	Container orchestration
ML Platform	Kubeflow	1.8	MLOps
Streaming	Apache Kafka	3.6	Event streaming
Stream Processing	Apache Flink	1.18	Real-time analytics
ETL	Apache NiFi	1.24	Data integration
Time-Series DB	InfluxDB	2.7	Sensor data storage
Relational DB	PostgreSQL	16	Metadata, configs
Vector DB	Milvus	2.3	RAG embeddings
Object Storage	MinIO	Latest	Files, models, backups
Cache	Redis	7.2	Session, query cache
ML Framework	PyTorch	2.1	Model training
CV Framework	DeepStream	6.3	Video analytics
LLM	Llama 3.1	70B	RAG generation
RAG Framework	LangChain	0.1	RAG pipeline
Model Serving	Triton	23.10	Model inference
API Framework	FastAPI	0.104	REST APIs
Frontend	React	18.2	Web dashboards
Mobile	React Native	0.72	Mobile app
Monitoring	Prometheus + Grafana	Latest	Observability
Logging	ELK Stack	8.11	Log management
Tracing	Jaeger	1.51	Distributed tracing
Security	Keycloak	22.0	IAM
Secrets	HashiCorp Vault	1.15	Secrets management

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9. APPENDIX

9.1 Glossary

(See BRD document)

9.2 Architecture Decision Records (ADRs)

ADR	Decision	Rationale
ADR-001	Edge-first architecture	Reduce latency, bandwidth, enable offline
ADR-002	Kubernetes for orchestration	Industry standard, vendor neutral
ADR-003	Kafka for event streaming	Proven at scale, replay capability
ADR-004	InfluxDB for time-series	Native ML integration, performance
ADR-005	Llama 3.1 for RAG	On-premise, fine-tunable, no data egress
ADR-006	Milvus for vector DB	Open-source, on-premise, scalable

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Document Approval

Role	Name	Signature	Date
Solution Architect			
IT Director			
Security Architect			

END OF HLD DOCUMENT